

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester IV)
MERN Stack

Practical - II

Roll No.: TY18	Name: Sachin Ramesh Mehta
Class: TYIT	Batch: 1
Date of Assignment: 30/06/2026	Date/Time of Submission: 30/06/2026

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code :-

```
function calculateSalary(name, basicSalary) {  
  let bonus = 5000;  
  let totalSalary = basicSalary + bonus;  
  
  console.log("Employee name:",name);  
  console.log("Basic Salary :",basicSalary);  
  console.log("Bonus :",bonus);  
  console.log("Total Salary :",totalSalary);  
  
}  
  
calculateSalary("Sachin",40000);  
console.log("Sachin Ramesh Mehta, TY18");
```

Output :-

```
[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2a.js"  
Employee name: Sachin  
Basic Salary : 40000  
Bonus : 5000  
Total Salary : 45000  
Sachin Ramesh Mehta, TY18  
  
[Done] exited with code=0 in 0.17 seconds
```

Code :-

```
function calculateSalary(product, quantity, price) {  
  let total = quantity*price;  
  let bill = total+(total*0.18);  
  
  console.log("Product :",product);  
  console.log( "Quantity :",quantity);  
  console.log("Price per Item :",price);  
  console.log("Total bill :",bill);  
  
}  
  
calculateSalary("Laptop",2,50000);  
  
console.log("Sachin Ramesh Mehta, TY18");
```

Output :-

```
[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2a.js"  
Product : Laptop  
Quantity : 2  
Price per Item : 50000  
Total bill : 118000  
Sachin Ramesh Mehta, TY18
```

b. Work with arrays and objects in JavaScript.

Code :-

```
let fruits = ["apple","banana","orange","grapes","mango"];  
let student = {  
  name : "Sachin",  
  age : 20,  
  grade : "A"  
}  
console.log("Fruits :",fruits);  
console.log("Fruits :",fruits[0]);  
console.log("Name :",student.name);  
console.log("Age :",student.age);  
console.log("Grade :",student.grade);  
console.log("Sachin Ramesh Mehta, TY18");
```

Output :-

```
[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2b.js"
Fruits : [ 'apple', 'banana', 'orange', 'grapes', 'mango' ]
Fruits : apple
Name : Sachin
Age : 20
Grade : A
Sachin Ramesh Mehta, TY18
```

c. Demonstrate use of arrow functions.

Code :-

```
let add = (a,b) => {
  return a+b;
}
let sub = (a,b) => {
  return a-b;
}
let mul = (a,b) => {
  return a*b;
}
let div = (a,b) => {
  return a/b;
}
console.log("Addition :",add(10,20));
console.log("Subtraction :",sub(10,20));
console.log("Multiplication :",mul(10,20));
console.log("Division :",div(10,20));
console.log("Sachin Ramesh Mehta, TY18");
```

Output :-

```
[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2c.js"
Addition : 30
Subtraction : -10
Multiplication : 200
Division : 0.5
Sachin Ramesh Mehta, TY18

[Done] exited with code=0 in 0.18 seconds
```

d. Write a program using array methods (map, filter, reduce).

Code :-

```
let numbers = [10,20,30,40,50];
let doubled = numbers.map(num => num*2);

let greater25 = numbers.filter(num => num > 25);

let total = numbers.reduce((sum,num) => sum + num,0);

console.log("Original array :",numbers);
console.log("Map :",doubled);
console.log("Filter :",greater25);
console.log("Reduce :",total);
console.log("Sachin Ramesh Mehta, TY18");
```

Output :-

```
[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2d.js"
Original array : [ 10, 20, 30, 40, 50 ]
Map : [ 20, 40, 60, 80, 100 ]
Filter : [ 30, 40, 50 ]
Reduce : 150
Sachin Ramesh Mehta, TY18

[Done] exited with code=0 in 0.227 seconds
```

e. Create a program that manipulates objects and displays output.

Code :-

```
let student = {
  rollNo: 101,
  name : "Sachin",
  marks : 85
}
//original data calling
console.log("Original Object :");
console.log(student);

//Accessing
console.log("\nAccessing Properties :");
console.log("Name :", student.name);
console.log("Marks :", student.marks);

//Updating
```

```
student.marks = 90;
console.log("\nAfter Updating marks :");
console.log(student);

//Adding
student.city = "Mumbai";
console.log("\nAfter adding city :");
console.log(student);

//deleting
delete student.rollNo;

console.log("\nAfter deleting Roll Number :");
console.log(student);

console.log("Sachin Ramesh Mehta, TY18");
```

Output :-

```
[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2e.js"
Original Object :
{ rollNo: 101, name: 'Sachin', marks: 85 }

Accessing Properties :
Name : Sachin
Marks : 85

After Updating marks :
{ rollNo: 101, name: 'Sachin', marks: 90 }

After adding city :
{ rollNo: 101, name: 'Sachin', marks: 90, city: 'Mumbai' }

After deleting Roll Number :
{ name: 'Sachin', marks: 90, city: 'Mumbai' }
Sachin Ramesh Mehta, TY18

[Done] exited with code=0 in 0.139 seconds
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code :-

```
//array of objects
let students = [
{
    rollNo: 101,
    name : "Sachin",
```

```

        marks : 85
    },
    {
        rollNo: 102,
        name : "Rahul",
        marks : 90
    },
    {
        rollNo: 103,
        name : "Rohit",
        marks : 95
    }
  ]];
//function to display Student details
function display(s){
  console.log("Student Details");
  console.log("-----");
  for(let stud of s)
  {
    console.log(`Roll No :${stud.rollNo}`);
    console.log(`Name :${stud.name}`);
    console.log(`Marks :${stud.marks}`);
    console.log(`-----`);
  }
}
//function call
display(students);
console.log("Sachin Ramesh Mehta, TY18");

```

Output :-

```

[Running] node "c:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical2f.js"
Student Details
-----
Roll No :101
Name :Sachin
Marks :85
-----
Roll No :102
Name :Rahul
Marks :90
-----
Roll No :103
Name :Rohit
Marks :95
-----
Sachin Ramesh Mehta, TY18
[Done] exited with code=0 in 0.188 seconds

```